

Topic to discuss

- Adams-Bashforth Method
- Numerical Problem
- Homework Problem

Adams-Bashforth Method

Another popular fourth order predictor-corrector method is the Adams-Bashforth-Moulton multistep method.

The predictor formula is also known as Adams-Bashforth predictor and is given by

$$y_{n+1} = y_n + \frac{h}{24} (55f_n - 59f_{n-1} + 37f_{n-2} - 9f_{n-3})$$

for $n=3$,

$$y_4 = y_3 + \frac{h}{24} (55f_3 - 59f_2 + 37f_1 - 9f_0)$$

And the corrector formula is known as Adams-Moulton corrector and is given by

$$y_{n+1} = y_n + \frac{h}{24} (f_{n-2} - 5f_{n-1} + 19f_n + 9f_{n+1})$$

for $n=3$,

$$y_4 = y_3 + \frac{h}{24} (f_1 - 5f_2 + 19f_3 + 9f_4)$$

* Note : To use this method, we need f_0, f_1, f_2, f_3 like Milne Predictor method.

An improved value of f_{n+1} is again corrected until it remains unchanged.

Numerical Problem :

Q: Use Adams-Bashforth Method, find $y(0.4)$

Given, $\frac{dy}{dx} = x + y$

$$y(0) = 1, \quad y(0.1) = 1.1103, \quad y(0.2) = 1.2428, \quad y(0.3) = 1.3997$$

Solution: We have,

$$x_0 = 0, \quad y_0 = 1 \quad \rightarrow \quad f_0 = 1$$

$$x_1 = 0.1, \quad y_1 = 1.1103 \quad \rightarrow \quad f_1 = 1.2103$$

$$x_2 = 0.2, \quad y_2 = 1.2428 \quad \rightarrow \quad f_2 = 1.4428$$

$$x_3 = 0.3, \quad y_3 = 1.3997 \quad \rightarrow \quad f_3 = 1.6997$$

Apply Adams-forth formula,

Predictor:

$$\begin{aligned}y_4 &= y_3 + \frac{h}{24} (55f_3 - 59f_2 + 37f_1 - 9f_0) \\&= 1.3997 + \frac{0.1}{24} [55(1.6997) - 59(1.4428) + 37(1.2103) - 9 \times 1] \\&= 1.5836\end{aligned}$$

Now, calculating, f_4

$$\begin{aligned}f_4 &= f(x_4, y_4) = f(0.4, 1.5836) \\&= 1.9836\end{aligned}$$

Corrector:

$$\begin{aligned} y_4^{(1)} &= y_3 + \frac{h}{24} [9f_4 + 19f_3 - 5f_2 + f_1] \\ &= 1.3997 + \frac{0.1}{24} [9(1.9836) + 19(1.6997) - 5(1.4428) + 1.2103] \\ &= 1.5836 \end{aligned}$$

Now again calculating f_4 ,

$$\begin{aligned} f_4 &= f(0.4, 1.5836) \\ &= 0.4 + 1.5836 \\ &= 1.9836 \end{aligned}$$

Again applying corrector method,

$$y_4^{(2)} = y_3 + \frac{h}{24} [9f_4 + 19f_3 - 5f_2 + f_1]$$

$$= 1.3997 + \frac{0.1}{24} [9(1.9836) + 19(1.6997) - 5(1.4428) + 1.2103]$$

$$= 1.5836$$

$$\therefore y_4^{(1)} \cong y_4^{(2)}$$

$$\text{So, } y_4 = y(0.4) = 1.5836$$

Ans.

Homework Problem :

Q: solve the ODE using Adams-Bashforth Methods

$$\frac{dy}{dx} = x - y, \quad y(0) = 1, \quad y(0.1) = 0.909675$$
$$y(0.2) = 0.837462, \quad y(0.3) = 0.781636$$

Solution :

$$\begin{aligned} x_0 &= 0, \quad y_0 = 1 && \rightarrow f_1 = -1 \\ x_1 &= 0.1, \quad y_1 = 0.909675 && \rightarrow f_2 = -0.809675 \\ x_2 &= 0.2, \quad y_2 = 0.837462 && \rightarrow f_3 = -0.637462 \\ x_3 &= 0.3, \quad y_3 = 0.781636 && \rightarrow f_4 = -0.481636 \end{aligned}$$

Predictor :

$$y_4 = y_3 + \frac{h}{24} (55f_3 - 59f_2 + 37f_1 - 9f_0)$$

After putting the value,

$$y_4 = 0.740646$$

Now calculating, f_4

$$f_4 = f(x_4, y_4) = f(0.4, 0.740646)$$

$$= -0.340646$$

Corrector ,

$$y_4^{(1)} = y_3 + \frac{h}{24} [9f_4 + 19f_3 - 5f_2 + f_1]$$
$$= 0.740639$$

Again correcting ,

$$f_4 = f(0.4, 0.740639)$$
$$= -0.340639$$

Again ,

$$y_4^{(2)} = y_3 + \frac{h}{24} [9f_4 + 19f_3 - 5f_2 + f_1]$$
$$= 0.740640$$

$\therefore$ We can see,

$$y_4^{(1)} \equiv y_4^{(2)}$$

So, $y(0.4) = 0.74064$ Ans.

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